Dragon Ball Miner × Quillon

Hardware Manufacturing Cooperation Proposal

Not Another ASIC — A New Product Category

Confidential Business Proposal

March 2026

QUG-V1 Pocket Supercomputer — 16-core RISC-V, 17 MH/s @ 9.5W, $149 retail

QUG-DOCK-16 — 16-slot enclosure, 272 MH/s @ 200W, each card is a full node

Quillon Forge — Dual EPYC 9755, liquid-cooled server, 4× RTX 5090 option

Quillon Foundation · https://quillon.xyz

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Executive Summary

This document proposes a **hardware manufacturing partnership** between Dragon Ball Miner and the Quillon Foundation. Dragon Ball Miner brings world-class chip fabrication relationships, PCB manufacturing, thermal engineering, and global distribution infrastructure. Quillon brings a production-ready quantum-resistant blockchain with a fully designed custom hardware platform.

> Dragon Ball Miner has the manufacturing pipeline; Quillon has the hardware design, the blockchain, and the market. Together, we can launch the **world's first RISC-V mining/node device** with a ready-made network of users.

Key points:

- Quillon's Proof-of-Work algorithm (BLAKE3 + 100-round VDF) is **fundamentally incompatible with traditional ASICs**. We are not asking Dragon Ball to build an ASIC.
- We **have** a complete hardware design — the QUG-V1 Pocket Supercomputer (see companion whitepaper) — that needs a manufacturing partner.
- Each QUG-V1 card is a **full blockchain node**, not just a hash accelerator. This is an entirely new product category for Dragon Ball.
- The **Quillon Forge** server platform (∼$24,500–$42,000) provides an enterprise-tier product line with GPU mining capabilities.
- Combined, these products span the entire market: $149 consumer cards → $2,550 DOCK enclosures → $42,000 enterprise servers.

> This proposal builds on two existing technical whitepapers:
> 1. **QUG-V1 Pocket Supercomputer** (v1.0.0) — Full SoC architecture, ISA extensions, thermal analysis, manufacturing BOM, and QUG-DOCK enclosure specifications. *See companion document:* `qug-v1-pocket-supercomputer.pdf`
> 2. **Q-NarwhalKnight Investor Pitch** (v2, February 2026) — Complete blockchain overview, tokenomics, competitive landscape, and self-reinforcing success dynamics. *See companion document:* `investor-pitch-v2.pdf`
>
> Both documents are provided alongside this proposal.

Why Traditional ASICs Cannot Mine QUG

Dragon Ball Miner has built highly successful ASICs for Kaspa (kHeavyHash), Alephium (Blake3), NEXA (NexaPow), and Radiant (SHA-512/256). These algorithms share a common property: **parallelism**. More hashing cores in parallel = more hashes per second = higher probability of finding a valid nonce.

Q-NarwhalKnight's mining algorithm is **fundamentally different**.

The BLAKE3 + 100-Round VDF Chain

QUG's Proof-of-Work requires evaluating a **Verifiable Delay Function (VDF)** chain:

1. Compute $h_0 = \text{BLAKE3}(\text{challenge} \,\|\, \text{nonce})$
2. Compute $h_1 = \text{BLAKE3}(h_0)$
3. Compute $h_2 = \text{BLAKE3}(h_1)$

   ⋮

4. Compute $h_{99} = \text{BLAKE3}(h_{98})$
5. If $h_{99} <$ target: valid solution

Each hash **depends on the previous hash**. This creates a **sequential chain of 100 BLAKE3 evaluations per nonce candidate**. The chain *cannot* be parallelized — step $k+1$ requires the output of step k.

A GPU with 10,000 CUDA cores can evaluate 10,000 *different nonces* in parallel — but each nonce still takes 100 sequential BLAKE3 hashes. A single fast core running at 1.5 GHz evaluates the same nonce chain in ~0.5 µs.
10,000 slow parallel cores ≈ 10,000 fast sequential cores.
There is no "ASIC advantage" from parallelism because the algorithm is inherently serial *per nonce*. The optimal hardware is many independent fast cores — which is exactly what the QUG-V1 already is.

Why This Matters for Dragon Ball

Traditional ASIC design focuses on maximizing hash throughput via massive parallelism (hundreds of hashing engines on one die). For QUG:

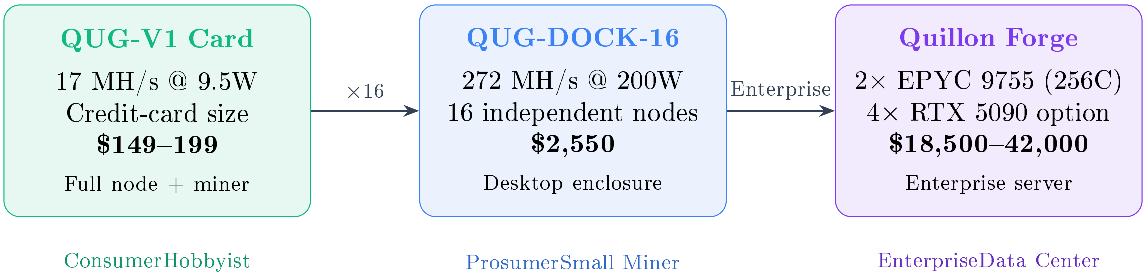
Traditional ASIC Approach	**Why It Fails for QUG**
Pack 256 SHA-256 engines on die	Each engine must still do 100 sequential BLAKE3 hashes per nonce — no throughput gain over 256 independent cores
Optimize hash pipeline width	VDF chain is inherently 32-byte wide (BLAKE3 output) — wider datapaths waste area
Remove CPU overhead (no OS)	QUG nodes *must* run a full blockchain node (P2P, storage, consensus) — a bare-metal hash chip cannot participate in the network

The Key Insight

The QUG-V1 chip design *already is* the optimal hardware for mining QUG. It is a 16-core RISC-V processor with custom `Xcrypto` BLAKE3 acceleration that achieves 9 cycles per hash. No ASIC redesign would meaningfully improve upon this because **the bottleneck is sequential latency, not parallel throughput**.
Dragon Ball's value is not in redesigning the chip — it is in **manufacturing** the chip we have already designed.

The Product Lineup: Three Tiers, One Network

Quillon has designed a **complete hardware product line** spanning consumer to enterprise:



Tier 1: QUG-V1 Pocket Supercomputer ($149–199)

The full technical specification is provided in the companion whitepaper (`qug-v1-pocket-supercomputer.pdf`). Key specifications:

Parameter	Value
Architecture	16-core heterogeneous RISC-V cluster (RV64GC)
Process node	TSMC N5 (5 nm FinFET)
Die size	∼50 mm^2
Mining cores	8× RV64 + `Xcrypto` BLAKE3 acceleration
Consensus cores	4× RV64 + RVV 1.0 (vector)
PQ-Crypto cores	2× RV64 + `Xlattice` (Dilithium5/Kyber1024 NTT)
I/O cores	2× RV64 + Ethernet MAC + TRNG
Mining throughput	17 MH/s (8 cores × 2.12 MH/s each)
Power consumption	9.5 W total (mining)
Energy efficiency	1.79 MH/s/W (163× better than RTX 4090)
Form factor	85 mm × 54 mm (credit card)
Interface	USB-C (power + Ethernet-over-USB)
SoC cost (10K vol.)	$25–35
Retail price	$149–199

Critical differentiator: Each QUG-V1 card runs a **complete Q-NarwhalKnight full node** — its own copy of the blockchain, its own libp2p peer identity, its own vote in DAG-Knight consensus. This is not a USB hash stick. It is a pocket supercomputer that mines *and* validates *and* routes transactions *and* participates in consensus.

Tier 2: QUG-DOCK-8 and QUG-DOCK-16 ($99–199 dock only)

Specification	QUG-DOCK-8	QUG-DOCK-16
Card slots	8	16
Mining throughput	136 MH/s	272 MH/s
Network nodes	8 independent	16 independent
Backplane	PCIe Gen3 ×4/slot	PCIe Gen3 ×4/slot
Network switch	Managed GbE (100 Mbps/slot)	Managed GbE (100 Mbps/slot)
Power (cards + dock)	91 W	167 W
Enclosure	Desktop form factor	Desktop form factor
Dock price (no cards)	$99–129	$149–199
Full system price	∼$1,300	∼$2,550

Throughput scales **linearly**: $H_{\text{dock}}(N) = N \times 17\,\text{MH/s}$, with zero inter-card coordination overhead. Each card mines on independent nonce ranges.

Tier 3: Quillon Forge Enterprise Server ($18,500–42,000)

The Quillon Forge is a liquid-cooled dual-socket EPYC server designed for enterprise mining, AI inference, and full-node operation. Full specification in `QUILLON_FORGE_MANUFACTURING_PIPELINE.md`.

Configuration	Specifications & Price
Base (CPU-only)	2× AMD EPYC 9755 (256 cores total), 512GB DDR5, 2×3.84TB NVMe, liquid-cooled — **$18,500**
GPU Standard	+ 2× NVIDIA RTX 5090 (2×32GB VRAM) — **$24,500**
GPU Pro	+ 4× NVIDIA RTX 5090 (4×32GB VRAM) — **$28,500**
AI Research	+ 2× NVIDIA A100 80GB — **$32,000**
AI Fleet	+ 4× NVIDIA L40 48GB — **$42,000**

GPU mining integration: As of March 2026, the Q-NarwhalKnight standalone miner and Slint desktop wallet both support OpenCL GPU mining with the same BLAKE3+VDF kernel. GPU-equipped Forge servers can mine with both CPU threads *and* GPU simultaneously, maximizing hardware utilization.

The standalone `q-miner` and the Slint wallet now both support multi-GPU OpenCL mining. Each RTX 5090 provides significant additional hashrate alongside the 256 CPU cores. The GPU Pro (4× RTX 5090) configuration is particularly compelling for operators who want maximum hashrate from a single managed server. The **GPU mining software is production-ready today** — the hardware just needs to be offered with more GPU options.

The Forge is tokenized as the **$FORGE RWA token** on Q-NarwhalKnight (500 total supply, burn-to-redeem for a physical machine). This creates a secondary market for Forge servers and integrates hardware sales directly into the blockchain ecosystem.

Where Dragon Ball Fits: Manufacturing Partner

Dragon Ball's Core Competencies

Dragon Ball Miner has demonstrated excellence in exactly the capabilities Quillon needs to bring the QUG-V1 to market:

Dragon Ball Capability	QUG-V1/DOCK Application
Chip fabrication relationships	TSMC/Samsung access for the 5 nm RISC-V SoC tape-out
PCB design & manufacturing	QUG-V1 credit-card PCB, DOCK backplane, edge connectors
Thermal engineering	9.5W passive cooling for card, 200W active cooling for DOCK-16
Enclosure design	DOCK-8/DOCK-16 desktop enclosures, fan/airflow optimization
Power delivery design	USB-C PD for single card, internal PSU for DOCK
Quality assurance & testing	Burn-in testing, MTBF validation, RoHS/CE/FCC compliance
Global distribution	Retail and wholesale channels to mining community worldwide
Volume pricing leverage	SoC cost reduction below $25 at Dragon Ball's order volumes

What Quillon Provides

Quillon Contribution	Details
Complete SoC design	RTL-ready 16-core RISC-V design with Xcrypto + Xlattice extensions
Firmware & OS	Bare-metal mining firmware + Linux full-node stack
Blockchain network	500,000+ LOC production blockchain with live network
Mining software	Production GPU + CPU miners (OpenCL, multi-device)
Wallet software	Desktop (Slint) + Web wallet with built-in miner
User base	Active mining community on quillon.xyz
Tokenized distribution	$FORGE RWA tokens for Forge servers; similar model available for QUG-V1

Not Another ASIC — A New Product Category

Every product Dragon Ball has shipped — the KAS miner, the ALPH miner, the NEXA miner — is a **dedicated hashing device**. It does one thing: compute hashes. It has no storage, no networking intelligence, no blockchain state.

The QUG-V1 is a **complete computer**. It runs a full operating system, stores its own copy of the blockchain, has its own cryptographic identity, participates in peer-to-peer networking, validates transactions, votes in consensus, *and* mines. Selling QUG-V1 cards is like selling miniature blockchain servers, not hash sticks.

The Market Positioning

	Traditional ASIC	GPU Miner	QUG-V1	Raspberry Pi
Mines	Yes	Yes	Yes	Barely
Full node	No	Optional	**Always**	Yes
Consensus vote	No	No	**Yes**	Yes
Peer identity	No	No	**Yes**	Yes
Form factor	Rack/shelf	Desktop	**Credit card**	SBC
Power	500–3000W	300–1000W	**9.5 W**	5–15W
QUG hashrate	N/A	∼5 MH/s	**17 MH/s**	∼200 kH/s
Price	$3,000–15,000	$1,500–5,000	**$149–199**	$80

Each QUG-V1 sold strengthens the network. Unlike traditional ASICs that centralize hashrate without contributing nodes, every QUG-V1 card adds a full peer to the decentralized network. This makes the product *aligned* with the blockchain's philosophy rather than adversarial to it.

Dragon Ball's New Customer Segment

Dragon Ball's current customers are industrial miners who buy rack-mounted ASIC units. The QUG-V1 opens a **new market segment**:

- **Individual hobbyists** who want a plug-and-play miner for $149
- **Crypto enthusiasts** who want to run a full node without managing a server
- **Small businesses** who want a DOCK-16 for quiet office mining
- **Existing Dragon Ball customers** who want to diversify into QUG mining alongside KAS/ALPH
- **Institutional operators** who want Quillon Forge servers with GPU mining

This is **additive** to Dragon Ball's existing business, not cannibalistic.

Economics and Revenue Model

QUG-V1 Card Economics

Component	Cost (10K vol.)	Share
QUG-V1 SoC (TSMC N5, 50 mm^2)	$25–35	45%
LPDDR5 2GB	$8–12	14%
eMMC 32GB + NOR 16MB	$6–9	10%
PCB (6-layer HDI, 85×54mm)	$4–6	7%
USB-C PD controller + connectors	$3–5	6%
Passive cooling (heatspreader)	$2–3	4%
Assembly & test	$5–8	10%
Packaging & shipping	$2–3	4%
Total BOM + Manufacturing	**$55–81**	100%
Retail price (2× markup)	$149–199	—
Gross margin	**52–59%**	—

QUG-DOCK Economics

Component	DOCK-8	DOCK-16
Managed Ethernet switch IC	\$15	\$15
Backplane PCB + edge connectors	\$12	\$20
Enclosure (aluminum, injection-molded)	\$15	\$25
PSU (120W / 250W)	\$10	\$18
Fans + thermal management	\$5	\$8
Controller MCU + firmware	\$4	\$4
Assembly & test	\$8	\$12
Total BOM	**\$69**	**\$102**
Retail price	\$99–129	\$149–199
Gross margin	30–47%	32–49%
Full system (cards + dock)		
Cards (8× or 16× @ \$149)	\$1,192	\$2,384
+ Dock	\$99–129	\$149–199
Full system retail	**∼\$1,300**	**∼\$2,550**

Revenue Projections (Conservative)

Product	Year 1 Units	ASP	Revenue	Gross Profit
QUG-V1 Card	10,000	\$175	\$1,750,000	\$962,500
QUG-DOCK-8	500	\$115	\$57,500	\$23,000
QUG-DOCK-16	300	\$175	\$52,500	\$22,050
Quillon Forge	50	\$26,000	\$1,300,000	\$598,000
Total Year 1			**\$3,160,000**	**\$1,605,550**

Proposed Cooperation Structure

Option A: Manufacturing Partnership

Dragon Ball manufactures the QUG-V1, DOCK, and Forge hardware under license from Quillon:

- **Dragon Ball role**: Tape-out management, PCB manufacturing, assembly, QA, distribution
- **Quillon role**: SoC RTL design, firmware, blockchain software, brand, community
- **Revenue split**: Negotiable (e.g., 60/40 or 70/30 in favor of manufacturer)
- **Branding**: Co-branded "Dragon Ball × Quillon" or Quillon brand with "Manufactured by Dragon Ball"

Option B: Joint Venture

Form a joint entity ("Quillon Hardware" or similar) that owns the hardware product line:

- Shared investment in tape-out costs (∼\$2–5M for TSMC N5 NRE)
- Shared ownership of manufactured inventory
- Dragon Ball contributes manufacturing infrastructure; Quillon contributes IP and software

- Profits shared per equity stake

Option C: OEM/ODM Arrangement

Dragon Ball produces QUG-V1 hardware as an OEM, selling units to Quillon at cost + margin:

- Simplest legal structure
- Dragon Ball earns manufacturing margin; Quillon earns retail margin
- Quillon handles all branding, sales, and distribution
- Dragon Ball can also sell through their own channels under co-brand

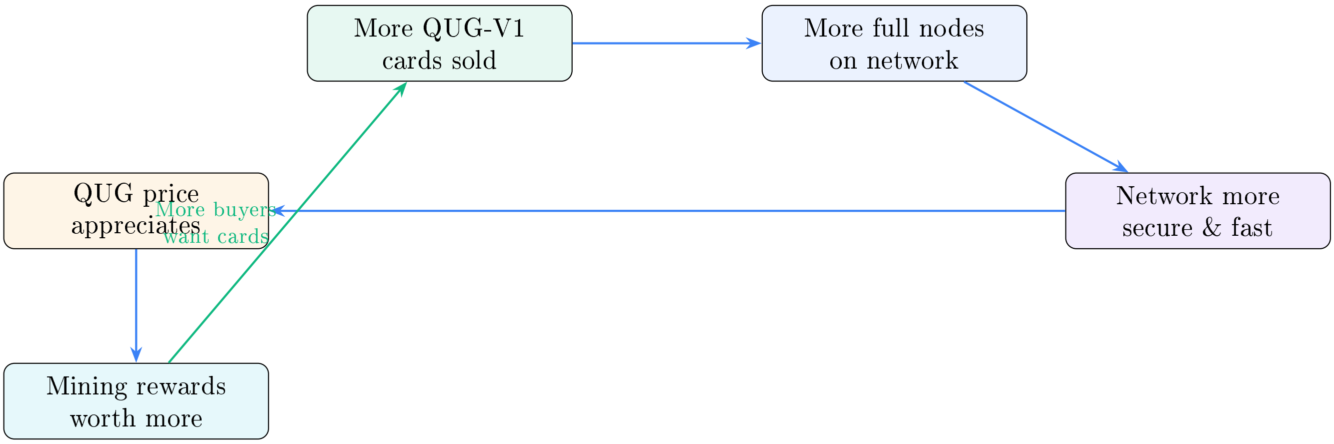
Development and Manufacturing Timeline

Phase	Milestone
2026 Q2	Partnership agreement signed. Joint review of QUG-V1 RTL design. Dragon Ball provides DFM (Design for Manufacturability) feedback.
2026 Q3	FPGA prototype validation. Xcrypto BLAKE3 extension verified at 100 MHz (∼1 MH/s). Firmware boots and mines on FPGA.
2026 Q4	PCB design finalized (credit-card form factor). DOCK-8/DOCK-16 enclosure prototyping.
2027 Q1	TSMC 7 nm tape-out (conservative first silicon). Target: 10 MH/s at 12 W.
2027 Q2	First silicon bring-up. Firmware porting from FPGA to silicon. Thermal validation.
2027 Q3	TSMC 5 nm tape-out (production die). Volume manufacturing begins. First $149 retail units ship. QUG-DOCK-8 and DOCK-16 ship alongside.
2027 Q4	Scale to 10,000+ units/quarter. Global retail distribution.

Note: The FPGA prototype phase (2026 Q3) can proceed immediately — it requires only FPGA development boards, not TSMC access. This allows validating the design and mining software before committing to a tape-out.

The Network Effect: Why Hardware Sales Compound

Unlike traditional ASIC sales (where the manufacturer's relationship with the customer ends at delivery), QUG-V1 sales create a **compounding network effect**:



Every card sold increases the value of every other card. This is the opposite of traditional ASIC economics, where more hashrate on the network *decreases* individual mining profitability. With QUG-V1, more cards = more nodes = stronger network = higher QUG value = more profitable mining for everyone.

Conclusion

Dragon Ball has the factory.
Quillon has the blueprint and the network.

Together, we launch the world's first RISC-V blockchain supercomputer in credit-card form factor.

Not an ASIC. Not a GPU. Something entirely new.

Next steps:

1. Review this proposal and the companion whitepapers
2. Schedule a technical deep-dive call to review the QUG-V1 RTL design
3. Sign mutual NDA for detailed IP sharing
4. Agree on cooperation structure (Option A, B, or C)
5. Begin FPGA prototype phase (can start immediately)

Website: https://quillon.xyz · **Live Network:** https://quillon.xyz

Companion Documents:
`qug-v1-pocket-supercomputer.pdf` — QUG-V1 Full Technical Whitepaper
`investor-pitch-v2.pdf` — Q-NarwhalKnight Investor Pitch & Technical Overview